

● NEURO360 — LIMITLESS BRAIN LAB

Staging Server Deployment Guide

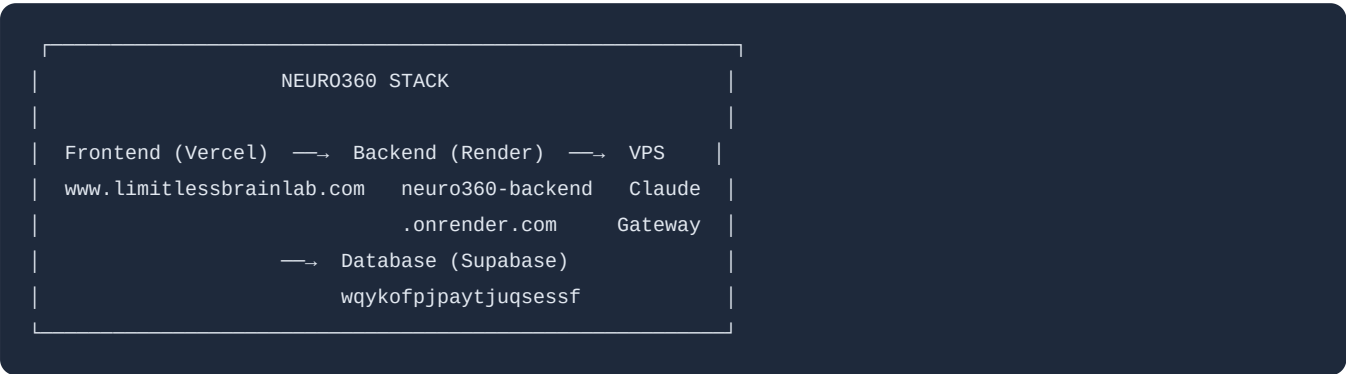
Complete manual for deploying the Neuro360 platform — Frontend,
Backend, Database, VPS Claude Gateway.

Version 1.0 | May 2026 | CONFIDENTIAL

Neuro360 — Manual Deployment Document

Version: 1.0 | Date: May 2026 | Project: Neuro360 / Limitless Brain Lab

Architecture Overview



Component	Platform	URL
Frontend	Vercel	www.limitlessbrainlab.com
Backend API	Render	neuro360-backend.onrender.com
Claude AI Gateway	Hostinger VPS	76.13.244.21:8787
Database	Supabase	wqykofpjpjpaytjuqsessf.supabase.co

Prerequisites — Accounts Required

Before deploying, you must have active accounts on:

- ☐ **GitHub** — Source code repository (chatgptnotes/Neuro360)
- ☐ **Vercel** — Frontend hosting (free tier)
- ☐ **Render** — Backend hosting (free tier)
- ☐ **Supabase** — Database + file storage (free tier)
- ☐ **Hostinger VPS** — Claude AI Gateway (VPS plan)
- ☐ **Stripe** — Payment processing (test + live keys)
- ☐ **Google Gmail** — Transactional email (App Password required)
- ☐ **Anthropic** — Claude API (for VPS gateway)

Part 1 — Database (Supabase)

1.1 Create Project

- Go to supabase.com → New Project
- Name: `neuro360`
- Region: Southeast Asia (Singapore)
- Save the following values:

- Project URL: `https://.supabase.co` - Anon Key (public) - Service Role Key (secret — never expose to frontend)

Run these in Supabase SQL Editor:

```
-- Core tables (run in order)
-- patients, clinics, organizations, reports, payments,
-- algorithmResults, clinical_reports, users
-- (Full schema available in /supabase/migrations/)
```

1.3 Storage Buckets

Create these buckets in Supabase Storage:

- `patient-reports` — patient PDF reports (public read)
- `neurosense-reports` — generated Claude reports
- `edf-files` — uploaded EEG files (private)

1.4 Row Level Security

Enable RLS on all tables. Patients can only read their own records.

Part 2 — VPS Claude Gateway (Hostinger)

The VPS runs the Nexaproc AI Gateway — it wraps the Claude CLI and provides PDF rendering via headless Chromium.

2.1 VPS Requirements

- Ubuntu 22.04 LTS
- Min 2GB RAM (4GB recommended)
- Node.js 20+
- PM2 (process manager)
- Chromium browser

2.2 Install Dependencies

```
# On the VPS (SSH in first)
curl -fsSL https://deb.nodesource.com/setup_20.x | sudo -E bash -
sudo apt-get install -y nodejs
sudo npm install -g pm2
sudo apt-get install -y chromium-browser
```

2.3 Install Claude CLI

```
npm install -g @anthropic-ai/claude-code
```

Login with your Anthropic account

```
claude login
```

2.4 Deploy Gateway Code

```
mkdir -p /opt/nexaproc-ai-gateway
cd /opt/nexaproc-ai-gateway
```

Copy gateway source files (bridge.ts, server.ts, t

```
npm install
npm run build
```

2.5 Create .env on VPS

```
cat > /opt/nexaproc-ai-gateway/.env << 'EOF'
PORT=8787
NEXAPROC_MASTER_KEY=<generate_a_strong_random_key_min_32_chars>
CLAUDE_BIN=claude
CLAUDE_TIMEOUT_MS=45000
CHROMIUM_BIN=chromium-browser
EOF
```

2.6 Create CLAUDE.md (AI Learning Memory)

```
cat > /opt/nexaproc-ai-gateway/CLAUDE.md << 'EOF'
```

Report Generation Lessons

Lesson: Blank pages in PDF

Lesson: HTML report templates must include `@page { size: A4 portrait; margin: 0 }` in CSS.

Lesson: Number extraction accuracy

Lesson: Copy values EXACTLY as printed. Never compute, round, or infer. Return null if absent.

Lesson: Narrative tone

Lesson: Use wellness/educational language only. Never write medical diagnoses.

EOF

2.7 Start with PM2

```
cd /opt/nexaproc-ai-gateway
pm2 start dist/server.js --name nexaproc-aiinmail
pm2 save
pm2 startup # follow the printed command to auto-start on reboot
```

2.8 Verify

```
curl http://localhost:8787/health
```

Expected: `{"status": "online", "engine": "Claude CLI"}`

2.9 Firewall — Open Port 8787

```
sudo ufw allow 8787/tcp
```

Part 3 — Backend API (Render)

3.1 Connect GitHub

1. Go to [render.com](#) → New Web Service
2. Connect GitHub → select `chatgptnotes/Neuro360`
3. Render will detect `render.yaml` and auto-configure both services

3.2 Backend Service Settings

Setting	Value
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3.3 Backend Environment Variables (Render Dashboard)

Set these in **Render Dashboard** → **neuro360-backend** → **Environment**:

Name	neuro360-backend	
Region	Singapore	
Plan	Free (or Starter for \$7/mo — recommended)	
Build Command	<code>cd server && npm install && npx puppeteer browsers install chrome</code>	
Start Command	<code>cd server && npm start</code>	
Health Check Path	<code>/api/health</code>	
Variable	Value	Notes

NODE_ENV	production	
PORT	3001	
SUPABASE_URL	https://.supabase.co	From Supabase
SUPABASE_SERVICE_ROLE_KEY	eyJ...	From Supabase (secret)
NEXAPROC_GATEWAY_URL	http://:8787	Your VPS IP
NEXAPROC_MASTER_KEY		Must match VPS .env
CLAUDE_REPORT_TOKEN		Static auth token
GEMINI_API_KEY	Aiza...	From Google AI Studio
GEMINI_REQUEST_DELAY_MS	2000	
GEMINI_DAILY_LIMIT	50	
STRIPE_SECRET_KEY	sk_live...	From Stripe Dashboard
STRIPE_WEBHOOK_SECRET	whsec...	From Stripe Webhooks
EMAIL_USER	noreply@yourdomain.com	Gmail address
EMAIL_PASS	xxxx xxxx xxxx xxxx	Gmail App Password
EMAIL_FROM	noreply@yourdomain.com	
EMAIL_TO	admin@yourdomain.com	Admin notification email
FRONTEND_URL	https://www.limitlessbrainlab.com	Your Vercel URL
SHARED_SSO_SECRET		
PUPPETEER_CACHE_DIR	/opt/render/project/src/.cache/puppeteer	

3.4 Disk Storage

Add a disk to the backend service:

- Name: uploads-storage
- Mount Path: /opt/render/project/src/server/uploads
- Size: 1 GB

3.5 Deploy

Click **Manual Deploy** → **Deploy Latest Commit**

3.6 Verify Backend

```
curl https://neuro360-backend.onrender.com/api/health
```

Expected: {"success":true,"message":"Server is running"}

Part 4 — Frontend (Vercel)

4.1 Import Project

1. Go to vercel.com → Add New Project
2. Import from GitHub → chatgptnotes/Neuro360
3. Framework: Vite (auto-detected)

4.2 Build Settings

Setting	Value
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4.3 Frontend Environment Variables (Vercel Dashboard)

Set these in **Vercel** → **Project Settings** → **Environment Variables**:

Build Command	<code>npm run build</code>
Output Directory	<code>dist</code>
Install Command	<code>npm install</code>
Variable	Value

<code>VITE_SUPABASE_URL</code>	<code>https://.supabase.co</code>
<code>VITE_SUPABASE_ANON_KEY</code>	<code>eyJ...</code> (anon key from Supabase)
<code>VITE_API_URL</code>	<code>https://neuro360-backend.onrender.com/api</code>
<code>VITE_CLAUDE_REPORT_TOKEN</code>	Same value as <code>CLAUDE_REPORT_TOKEN</code> on Render
<code>VITE_STRIPE_PUBLISHABLE_KEY</code>	<code>pk_live...</code> from Stripe
<code>VITE_APP_NAME</code>	<code>Neuro360</code>
<code>VITE_SUPABASE_STORAGE_BUCKET</code>	<code>patient-reports</code>

4.4 Custom Domain

1. Vercel → Project → Settings → Domains
2. Add `www.limitlessbrainlab.com`
3. Update DNS at your domain registrar:
 - CNAME `www` → `cname.vercel-dns.com`

4.5 Deploy

Push to `main` branch — Vercel auto-deploys on every push.

4.6 Verify Frontend

Open `https://www.limitlessbrainlab.com` — login page should load.

Part 5 — Stripe Webhooks

5.1 Create Webhook

1. Stripe Dashboard → Developers → Webhooks → Add Endpoint
2. URL: `https://neuro360-backend.onrender.com/api/stripe/webhook`
3. Events to listen for:
 - `checkout.session.completed` - `payment_intent.succeeded`

5.2 Copy Webhook Secret

Copy the `whsec...` secret and add it to Render as `STRIPE_WEBHOOK_SECRET`.

Part 6 — Gmail App Password (Email)

1. Google Account → Security → 2-Step Verification → App Passwords
2. App: Mail | Device: Other — name it "Neuro360"
3. Copy the 16-character password

Part 7 — Post-Deployment Checklist

Run through this after every deployment:

Infrastructure

- ☐ VPS gateway responds: `curl http://:8787/health` → `{"status":"online"}`
- ☐ Backend healthy: `curl https://neuro360-backend.onrender.com/api/health` → `{"success":true}`
- ☐ Frontend loads: open `https://www.limitlessbrainlab.com`

Authentication

- ☐ Can create a new account (Register)
- ☐ Can login with existing account
- ☐ Admin login works
- ☐ Clinic login works

Report Generation

- ☐ Upload EEG files → QEEG report generates
- ☐ "Upload to Claude" → sidecar health check shows `✔ Sidecar online`
- ☐ Claude report generates (12 pages, no blank pages)
- ☐ Report appears in patient portal under "Neurosense Reports"

Payments

- ☐ Stripe checkout opens (test mode)
- ☐ Webhook receives `checkout.session.completed`
- ☐ Credits update in `clinics` and `organizations` tables

Email

- ☐ Test email sends from admin panel

Part 8 — Environment Variables Quick Reference

Critical — Must Set Manually in Render Dashboard

<code>NEXAPROC_MASTER_KEY</code>	← Must match VPS <code>.env</code> exactly
<code>CLAUDE_REPORT_TOKEN</code>	← Must match <code>VITE_CLAUDE_REPORT_TOKEN</code> in Vercel
<code>SUPABASE_SERVICE_ROLE_KEY</code>	← Never expose to frontend
<code>STRIPE_SECRET_KEY</code>	← Use <code>sk_live_</code> for production

Critical — Must Set Manually in Vercel Dashboard

<code>VITE_CLAUDE_REPORT_TOKEN</code>	← Must match <code>CLAUDE_REPORT_TOKEN</code> on Render
<code>VITE_SUPABASE_ANON_KEY</code>	← Public key (safe for frontend)
<code>VITE_API_URL</code>	← Must point to Render backend URL

Part 9 — Common Deployment Issues

Problem	Cause	Fix
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Blank pages in PDF	Chromium paper size mismatch	Already fixed via <code>@page { size: A4 }</code> in template
"Sidecar offline" error	VPS gateway not running	<code>pm2 restart nexaproc-ainmail</code> on VPS
"NEXAPROC_MASTER_KEY not set"	Missing in Render dashboard	Add secret in Render → Environment
Backend cold start slow	Render free tier spins down	Upgrade to Starter (\$7/mo) or keep-warm cron
Report credits not updating	<code>clinics</code> and <code>organizations</code> out of sync	Update both tables in Supabase
Checkout fails	Stripe key missing or wrong	Verify <code>STRIPE_SECRET_KEY</code> in Render dashboard

Part 10 — Keeping the VPS Alive (Keep-Warm)

Render free tier spins down after 15 min idle. Add a keep-warm cron on the VPS:

```
# Add to crontab (runs every 10 minutes)
crontab -e

Add this line:

/10    curl -s https://neuro360-backend.onrender.com/api/health > /dev/null
```